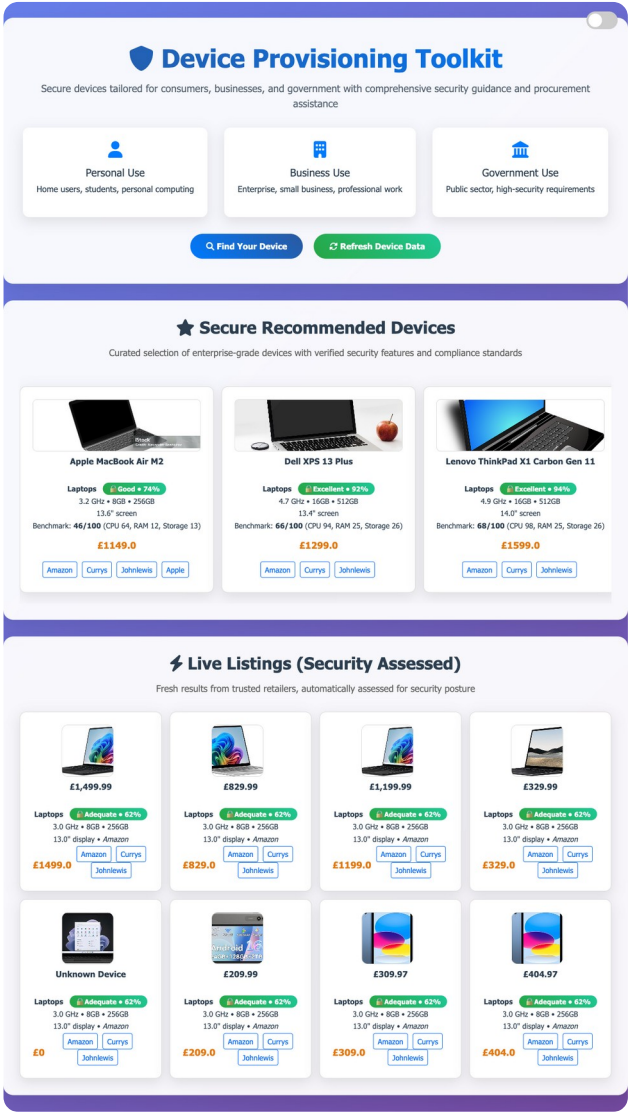


How do we decide what devices we provide?

A security-focused comparison and provisioning toolkit built from the dissertation.



The buying decision was missing a security standard



**FACULTY OF SCIENCE, ENGINEERING
AND COMPUTING**

School of Computer Science and Mathematics

**BSc (Hons) DEGREE
IN**

Cyber Security and Digital Forensics
Nathan Brown-Bennett

"How do we decide what devices we provide?"
18th July 2024

David Pickard

Kingston University London

The dissertation examined how governments, schools, employers and consumers choose devices—and whether security, practicality and reliability are considered together.

The gap

Guidance existed in fragments. Buyers could compare brand, price and specifications, but had no single workflow that translated security research into a usable decision.

Source: BSc dissertation, 18 July 2024

Three gaps kept recurring across the research



Security arrives late

Device decisions often begin with price, brand and familiarity; security is added after selection.



Guidance is fragmented

Hardware risks, software hardening and user capability are rarely explained in one place.



People need usable advice

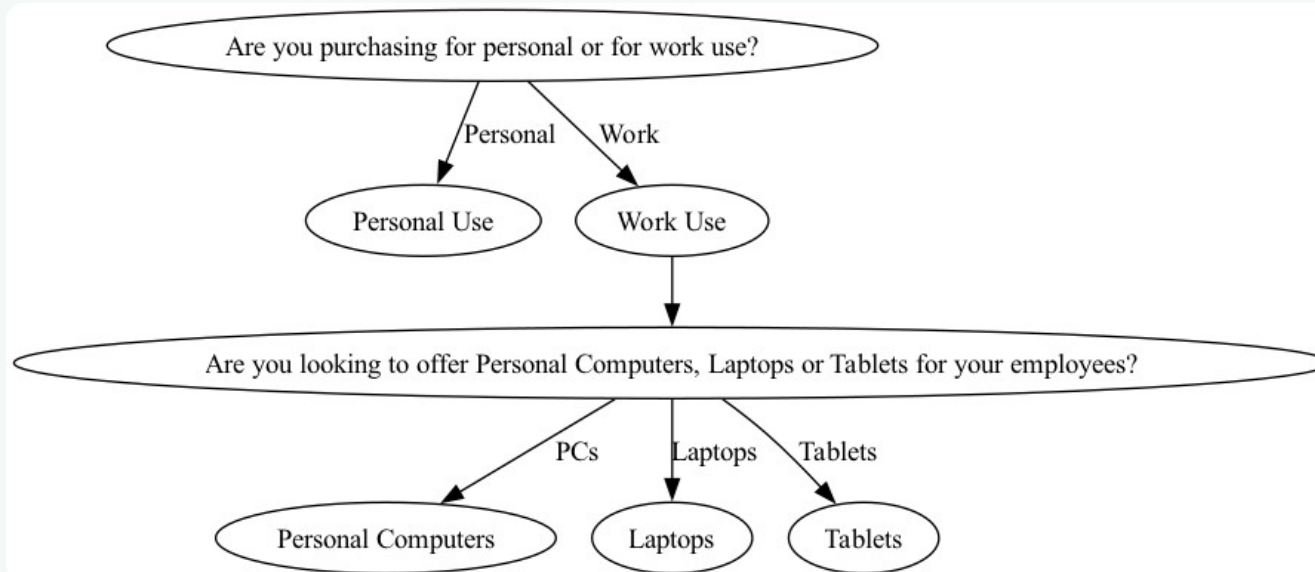
A technically correct recommendation still fails if buyers cannot understand or apply it.

The product opportunity was not another comparison list. It was a decision tool that could connect research to action.

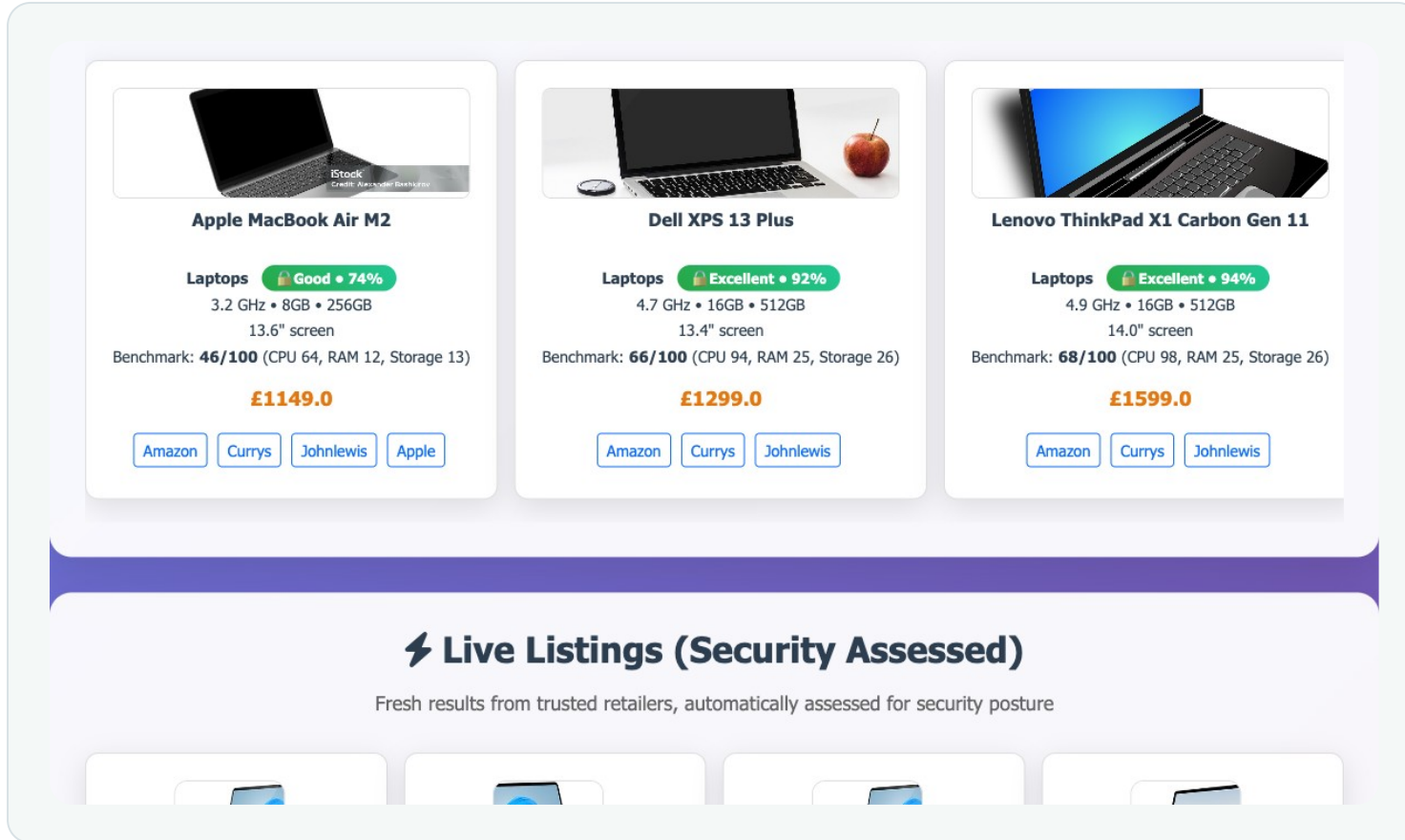
Research became a repeatable provisioning framework

The framework connects

- Buyer and user needs
- Hardware security considerations
- Operating-system and software controls
- Practicality and maintainability
- Clear recommendations



The current toolkit makes the research usable



The screenshot displays a web application interface for laptop listings. It features three main product cards, each with a laptop image, title, specifications, a security assessment badge, benchmark score, price, and retailer links. Below the cards is a purple banner with the text 'Live Listings (Security Assessed)' and a subtext 'Fresh results from trusted retailers, automatically assessed for security posture'. At the bottom, there are four small, partially visible laptop images.

Product	Assessment	Benchmark	Price	Retailers
Apple MacBook Air M2	Good • 74%	46/100 (CPU 64, RAM 12, Storage 13)	£1149.0	Amazon, Currys, Johnlewis, Apple
Dell XPS 13 Plus	Excellent • 92%	66/100 (CPU 94, RAM 25, Storage 26)	£1299.0	Amazon, Currys, Johnlewis
Lenovo ThinkPad X1 Carbon Gen 11	Excellent • 94%	68/100 (CPU 98, RAM 25, Storage 26)	£1599.0	Amazon, Currys, Johnlewis

⚡ Live Listings (Security Assessed)
Fresh results from trusted retailers, automatically assessed for security posture

What exists now

A working Flask application with device discovery, structured comparisons, security guidance, purchase context and provisioning recommendations.

Python • Flask • SQLite • Graphviz • Pandas

Search starts with the person, not the specification sheet

★ Secure Recommended Devices

Curated selection of enterprise-grade devices with verified security features and compliance standards



Apple MacBook Air M2

Laptops

Good • 74%

3.2 GHz • 8GB • 256GB

13.6" screen

Benchmark: 46/100 (CPU 64, RAM 12, Storage 13)

£1149.0

Amazon

Currys

Johnlewis

Apple



Dell XPS 13 Plus

Laptops

Excellent • 92%

4.7 GHz • 16GB • 512GB

13.4" screen

Benchmark: 66/100 (CPU 94, RAM 25, Storage 26)

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Lenovo ThinkPad X1 Carbon Gen 11

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⚡ Live Listings (Security Assessed)

Fresh results from trusted retailers, automatically assessed for security posture



£1,499.99



£829.99



£1,199.99



£329.99

Browse and compare

FILE NAME

FILTERED DATA

Any Brand

£100

£1500

Show Advanced Filters

Find Secure Devices

Close

★ Secure Recommended Devices

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Describe the real use case

A device page explains the decision—not just the device



Dell XPS 13 Plus

Laptops • Windows 11 • Unknown

£1299.0

Excellent (92%) Enterprise Ready Latest Hardware

OverviewSecurity GuideComparePurchase

Performance Specifications

Processor: 4.7 GHz High Performance

Memory: 16 GB RAM Professional

Storage: 512 GB SSD Ample Space

Display: 13.4" Screen Standard

Why This Device?

- Excellent multitasking with 16GB+ RAM
- High-performance processor for demanding tasks
- Ample storage with fast SSD technology
- Latest generation hardware
- Enterprise-grade security features
- Security score: **92%** (Excellent)
- Compatible with modern business software

Hardware Benchmark Summary

Benchmark data unavailable for this device.

Back to SearchPrint GuideDownload Security Checklist

Specifications and score



Dell XPS 13 Plus

Laptops • Windows 11 • Unknown

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OverviewSecurity GuideComparePurchase

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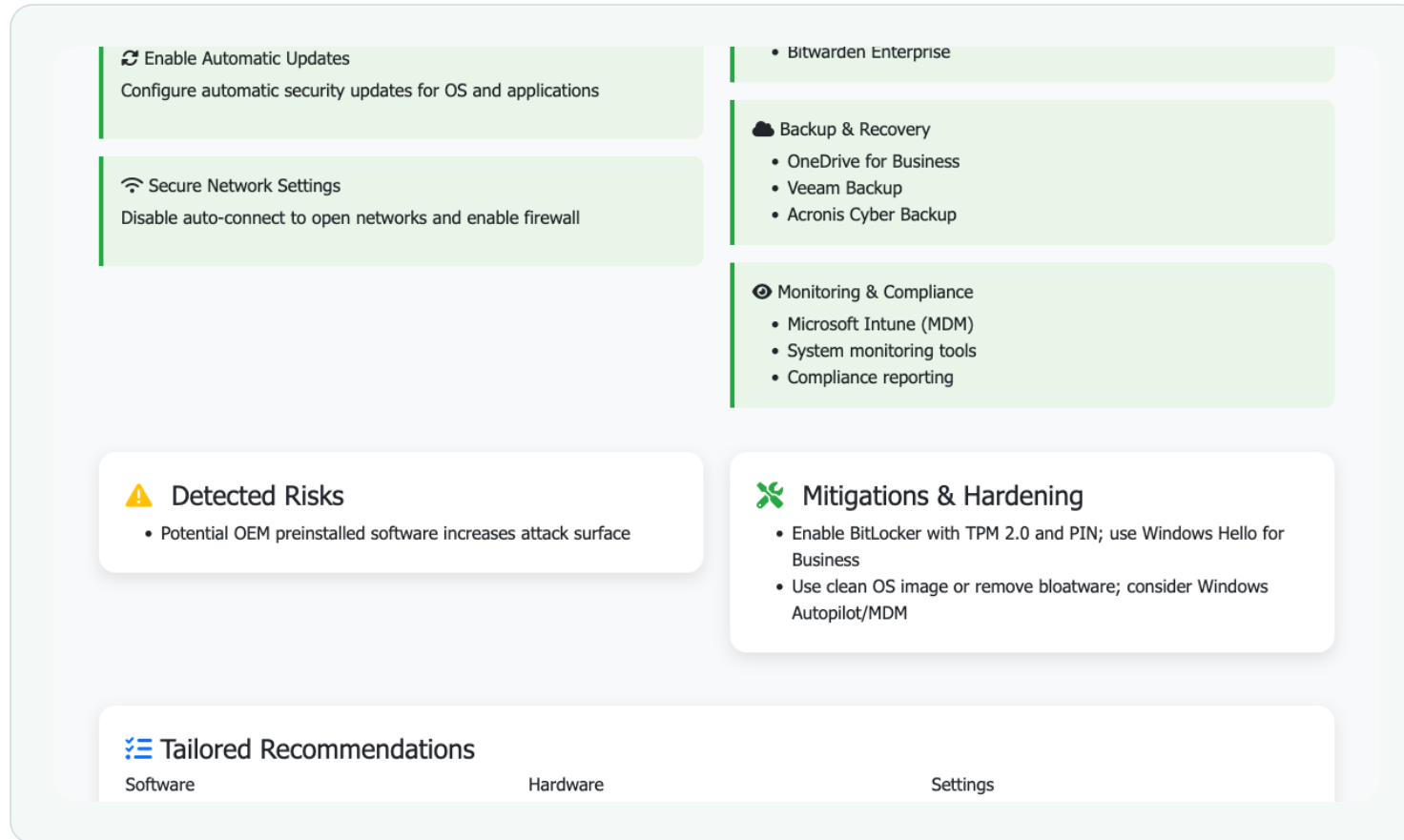
Hardware Benchmark Summary

Benchmark data unavailable for this device.

Back to SearchPrint GuideDownload Security Checklist

Security context and recommendation

The output turns a purchase into practical next steps



Beyond comparison

The toolkit gives buyers and administrators understandable guidance for setup, hardening and ongoing use—so security is part of the workflow rather than an afterthought.

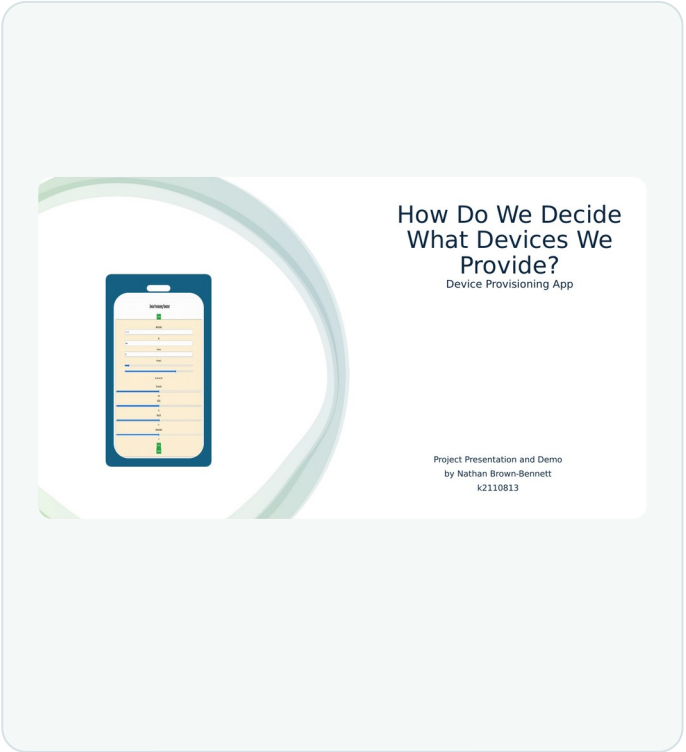
Research → recommendation → action

The strongest evidence is the progression from question to product



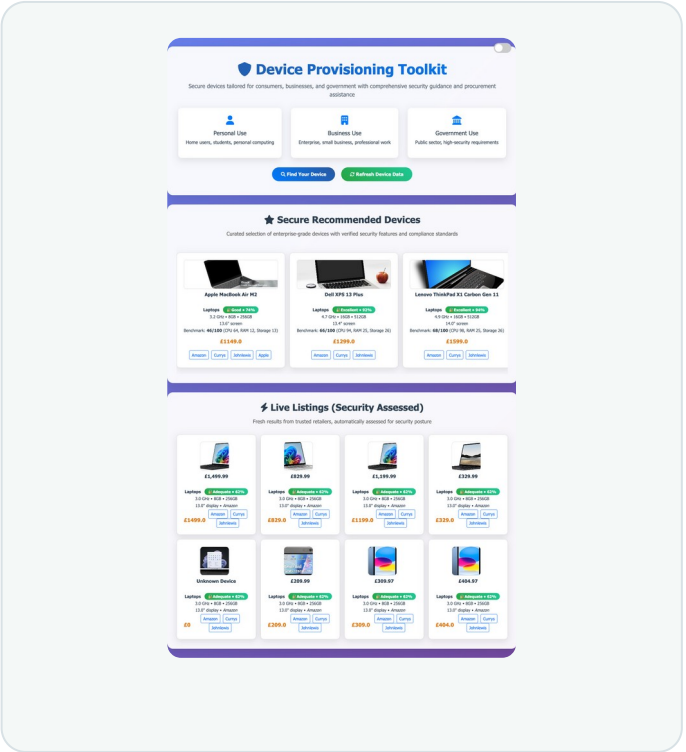
2024

Dissertation



2024

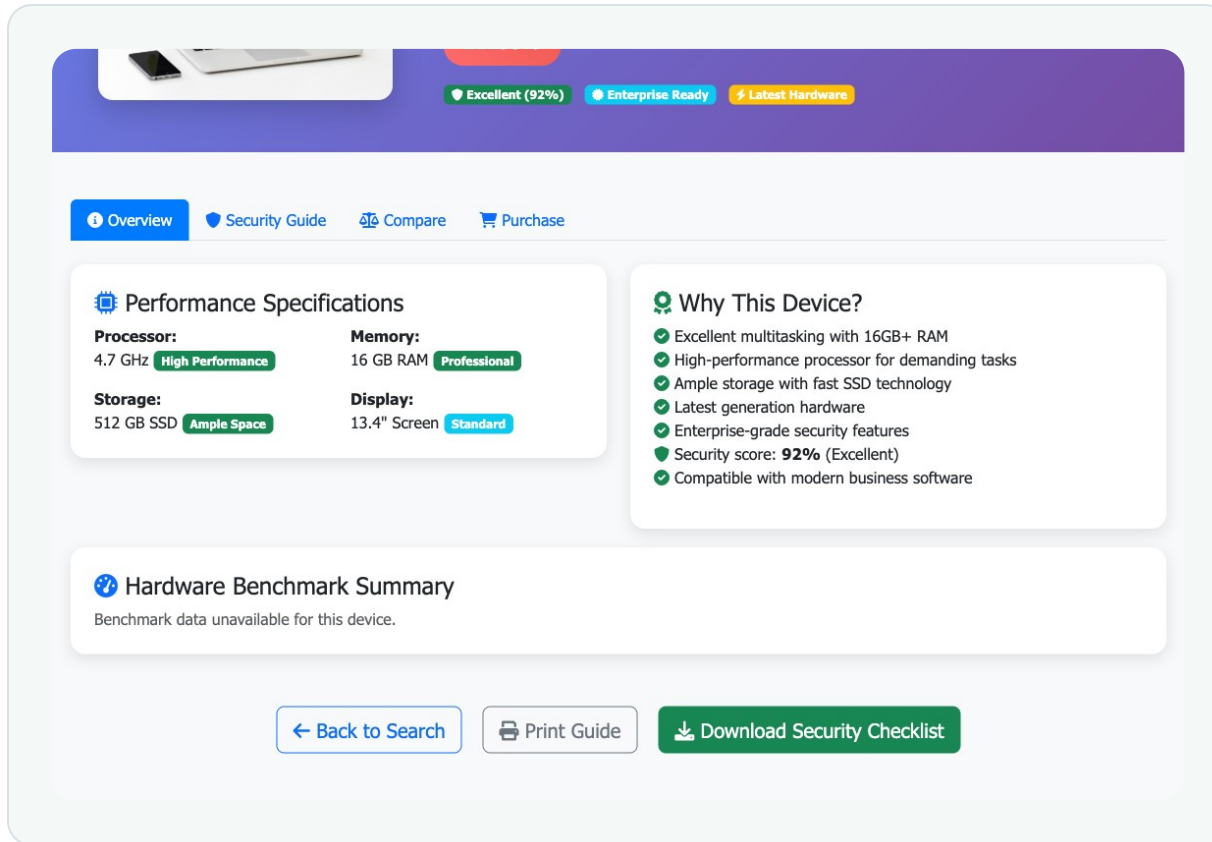
Original presentation



Current

Working toolkit

A lightweight stack keeps the working prototype portable



Product architecture

Python + Flask
Application and routing

SQLite + Pandas
Structured device evidence

Graphviz
Decision and provisioning flows

The next phase is stronger evidence, scoring and organisational validation




Broader evidence

Expand and maintain the device dataset with clearer provenance.



Explainable scoring

Make every recommendation traceable to transparent criteria.



Operational validation

Test workflows with procurement, IT support and security teams.

The product is already useful as a demonstrator; the opportunity is to make its evidence and operation durable enough for real organisations.

I build what should exist

This project started with a missing standard and became a working tool that makes device security decisions easier to understand and act on.

Dissertation

nathanbrownbennett.github.io/NathanBrown-Bennett/assets/documents/device-provisioning-dissertation.pdf

Repository

github.com/NathanBrownBennett/Dynamic-Flowchart-For-Device-Provisioning-

